

Vivek Vijayan

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PROFILES

 github.com/vivekvjyn
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EDUCATION

Master in Sound and Music Computing

UNIVERSITAT POMPEU FABRA

Barcelona, Spain | 2024 – 2026

Bachelor of Technology in Computer Science

KERALA TECHNOLOGICAL UNIVERSITY

Kerala, India | 2020 – 2024

Higher Secondary Education in Computer Science

ST. BERCHMAN'S HIGHER SECONDARY SCHOOL

Kerala, India | 2018 – 2020

WORK EXPERIENCE

PITCH INNOVATIONS | RESEARCH & DEVELOPMENT ENGINEER

Chennai, India | 2026 – Present

- Contributions mainly to Fluid Pitch 2, Sonic Atlas etc.
- Designing and developing the Carnatic-Engine which synthesizes Gamakas from plain Musical Instrument Digital Interface (MIDI), which can then be mapped to MIDI Polyphonic Expression (MPE).
- Designing and developing Neural Pitch Editor module for Sonic Atlas, which lets you edit instrument samples and make it your own.

MUSIC TECHNOLOGY GROUP | RESEARCH ASSISTANT

Barcelona, Spain | 2025 – 2026

- Assisting research in computational melodic analysis of Carnatic music.
- Improving and extending the *comPIAM* repository for computational analysis of Carnatic and Hindustani music.
- Contributed to *mirdata*, a dataset loader for widely used Music Information Retrieval (MIR) datasets.

FREELANCE | MUSIC PRODUCER

Kerala, India | 2019 – Present

- Worked as an additional programmer for feature films and advertisements.
- Composed original scores for short films; awarded **Best Music Director** at SMIFA Shorts 2019 (Malayalam category).
- Produced, mixed, and released music as an independent artist.
- Provided audio production services for cultural events.

PROJECTS

COCONUT AUDIO

C++, JUCE, CMAKE

Free and open-source VST plugin development initiative. Plugins include Old Guitarist (physically modelled guitar), Intellectual Gate (drum bleed removal), Pulp Fiction (multi-FX rack) and Sound of Music (multiband bit crusher).

MIRDATA

PYTHON

Open-source Python library for loading and exploring music/audio research datasets. Contributions: dataset loaders for the Saraga Audiovisual and IAM Melodic Similarity datasets.

COMPIAM

PYTHON, PYTORCH

Repository of datasets, tools and models for computational analysis of Indian Art Music. Contributions: TCN Carnatic and BeatThis Carnatic beat-tracking modules.

COMPUTATIONAL IAM TUTORIAL

PYTHON, JUPYTER

Contributed notebooks to the ISMIR 2022 tutorial on corpus-based research on Indian Art Music, covering the Saraga Audiovisual dataset and the TCN Carnatic module.

GEOSYNTH

Sonification of geolocations using an autoencoder, with data collected through the Freesound API.

PYTHON, TENSORFLOW, FLASK, DOCKER

MARKRETHE

Python package for logging experiments to a Markdown file.

PYTHON

TINKERHUB CEC WEBSITE

Official website of TinkerHub CEC, the TinkerHub chapter at College of Engineering, Chengannur.

JAVASCRIPT, REACT

PUBLICATIONS

Leveraging Melodic Context for Improved Svara Representation

2025

17TH INTERNATIONAL SYMPOSIUM ON COMPUTER MUSIC MULTIDISCIPLINARY RESEARCH

<https://zenodo.org/records/17497901>

A Raga Independent Encoder for Svara Representation in Carnatic Music

2026

13TH INTERNATIONAL CONFERENCE ON DIGITAL LIBRARIES FOR MUSICOLOGY

<https://dl.acm.org/doi/10.1145/3815723.3815730>

Adapting Deep Learning-based Meter Tracking to Carnatic Music

2026

23RD SOUND AND MUSIC COMPUTING CONFERENCE AND WINTER SCHOOL Accepted, to be presented

ACHIEVEMENTS

Best Paper Award

2025

17TH INTERNATIONAL SYMPOSIUM ON COMPUTER MUSIC MULTIDISCIPLINARY RESEARCH

AICTE Tuition Fee Waiver | ALL INDIA COUNCIL FOR TECHNICAL EDUCATION

2020

Best Music Director (Malayalam Category)

2019

STONED MONKEY INTERNATIONAL FILM AWARDS FOR SHORTS

VOLUNTEERING

IHRD THARANG 2023 | JOINT CONVENOR, MEDIA TEAM LEAD

2023

ALUMNI RELATION CELL, CEC

2021 – 2023

CONVOCATION CEREMONY COORDINATOR, CEC TALKS COORDINATOR

IEEE SB CEC | MEDIA TEAM MEMBER

2021 – 2023

TINKERHUB CE CHENGANNUR | MEDIA TEAM MEMBER

2021 – 2023

CECIFY RADIO | AUDIO EDITOR

2021 – 2023

NATIONAL SERVICE SCHEME | SOCIAL SERVICE VOLUNTEER

2018 – 2020

NATIONAL CADET CORPS | CADET (ARMY WING)

2015 – 2017

SKILLS

Programming: C++, Python, JavaScript, Rust, Git

Media: FL Studio, Ableton Live, Cubase, Reaper, RX, Photoshop, Premiere Pro, After Effects, Figma